

Foaming of amorphous polymers and blends in supercritical CO₂: Solubility versus block copolymers addition

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Abstract

Supercritical CO₂ (scCO₂) is used as a medium for foaming amorphous polymers. A study of the solubility of supercritical CO₂ in several amorphous polymers (PS and PMMA) and blends is performed, followed by an investigation of the foaming behavior of the polymer–gas systems. Nano-structuring triblock copolymers (styrene-co-butadiene-co-methylmethacrylate SBM and methylmethacrylate-co-butylacrylate-co-methylmethacrylate MAM) were blended as additives to PS or PMMA by extrusion. The addition of these triblock copolymers results in an important weight gain ratio of gas in a wide range of temperatures (from 25 to 80°C), relating this weight gain ratio to the foaming behavior of the blends (CO₂ is preferentially located in the micro or nano-domains issued from the structuration of the block copolymer). Foaming is carried out in a batch one-step scCO₂ process, keeping constant the saturation pressure and depressurization rate (300 bar and 60 bar/min, respectively). Influence of saturation temperature (25–85 °C) on the final porous structure is shown. In spite of the influence of the terpolymer on the weight gain ratio, the structuration is believed to provide a good control of microcellular foams in PS and PMMA.

Keywords

microcellular foaming, solubility, amorphous polymers, supercritical CO₂, block copolymers

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Introduction

Microcellular foams are defined as having average cell sizes in the range of 1–10 μm , and cell densities on the order of 10^9 – 10^{15} cells/cm³. Such structures were proposed by Suh et al.,¹ and it is well known that inert gases such as CO₂ can be used to foam amorphous polymeric materials. The low critical conditions of scCO₂ (31.1 °C and 73.8 bar) offer many advantageous properties, like a tuneable solvent power, plasticization of glassy polymers and higher diffusion rates.^{2,3} Following this line of investigation, the use of supercritical CO₂ as a physical foaming agent is investigated in view of enlarging the accessible cell-size range in amorphous commodity polymers (PS, PMMA) and polymer blends, towards ultramicro-cellular or nano-cellular structures. Cellular structures, in any gas process, result from a balance between several variables:

- (i) Diffusion of CO₂ in the polymer (accessibility).
- (ii) Affinity to CO₂ (solubility).
- (iii) Polymeric viscoelastic behavior (deformation, viscosities) and the glass-to-rubber transition (T_g , vitrification temperature).

To produce microcellular materials, a batch single step-process is employed. In this process, the polymer is saturated with CO₂ in the supercritical regime, for a fixed time. After saturation, the sample, approaching or being in a rubbery state, is depressurized to atmospheric pressure, taking advantage of the swelling and plasticization of the polymer, which reduces the glass transition temperature, allowing the gas expansion. In this method, the cellular microstructure may be controlled by changing the saturation temperature and depressurization rate. Polystyrene (PS),^{4,5} polycarbonate,⁶ polystyrene-co-methylmethacrylate,⁷ and other polymers such as polymethyl-methacrylate (PMMA)⁸ or polyethylene⁹ have been already obtained as microcellular polymers. The objective of the present study is to investigate the solubility of scCO₂ into amorphous polymers (here PS and PMMA) and blends over a wide range of temperatures (from 25 to 80°C). Moreover, the use of block copolymer additives will be explored in order to analyse their influence in the CO₂ solubility and the foaming behavior. The blends consist of a low content (10 wt%) triblock copolymer able to induce phase separation at a nano-meter or submicrometer level.

Materials

Two neat homopolymers were used. A transparent polymethylmethacrylate (PMMA, $M_w = 70,000$ g/mol) was kindly supplied by Altuglas Company-Arkema, in the form of pellets, together with high molar mass transparent polystyrene (PS, $M_w = 140,000$ g/mol), which was kindly supplied by Total Petrochemicals company, in the form of pellets. Moreover, two block copolymers were used as additives in blends with the previous polymers, supplied by Arkema Company, in the form of

Table 1. Some characteristics of the materials employed

Material	$T_{g0}(^{\circ}\text{C})$	$M_w(\text{g/mol})$	Bulk density $\rho_s(\text{g/cm}^3)$
PS	103	140,000	1.05
PMMA	111	70,000	1.17
PS + SBM	98	30,000	1.01
PMMA + MAM	109	80,000	1.02

pellets, and employed as received. These additives were one triblock styrene-butadiene-methylmethacrylate terpolymer (polystyrene-block-polybutadiene-block-polymethylmethacrylate named **SBM**), with a composition of 52 wt% styrene, 30 wt% butadiene, and 18 wt% methylmethacrylate with a PS block of molar mass equal to 30,000 g/mol. The other additive employed was one triblock methylmethacrylate-butylacrylate-methylmethacrylate terpolymer (poly methylmethacrylate-block-polybutylacrylate-block-polymethylmethacrylate named **MAM**) with 35 wt% methacrylate, 30 wt% butylacrylate and 35 wt% methacrylate, and a PMMA block of molar mass equal to 90,000 g/mol. Two blends were produced: first, a PS + SBM blend, with 90 wt% PS and 10 wt% SBM, second, a PMMA + MAM blend, with 90 wt% PMMA and 10 wt% MAM. In both cases, blending was carried out by extrusion and injection moulding was used to obtain the bulk specimens in the form of parts of $15 \times 60 \text{ mm}^2$ with 1.5 mm thickness. Blends of PS + MAM and PMMA + SBM were also produced, but due to the complete immiscibility with the homopolymers, samples obtained were phase-separated at a macro-scale level, showing no transparency with a heterogeneous structure. For this reason, these materials were rejected for micro foaming proposes. In Table 1 are shown some physical properties of all the materials.

Experimental

Microcellular foaming experiments were carried out in a high pressure reactor provided by TOP Industry (France), with a capacity of 300 cm^3 and capable of operating at 250°C and 400 bar. The reactor is equipped with a pressure pump controller provided by Teledyne ISCO, and controlled to maintain automatically the temperature and pressure at given values. The CO_2 vessel temperature and pressure were monitored in the course of the whole process. In our procedure, depressurization time was fixed, varying the saturation temperature. In this procedure, samples were saturated with supercritical CO_2 at a maintained pressure of 300 bar during 24 h. Saturation temperatures varied between 25 and 80°C . It is important to notice that the weight of gas absorbed between 16 and 24 h remains invariable in our samples (1.5 mm thickness). Moreover, in our previous works, other processing parameters such as different pressures were employed to analyze the effect of the saturation pressure in the cellular structure of the foamed materials.¹⁰

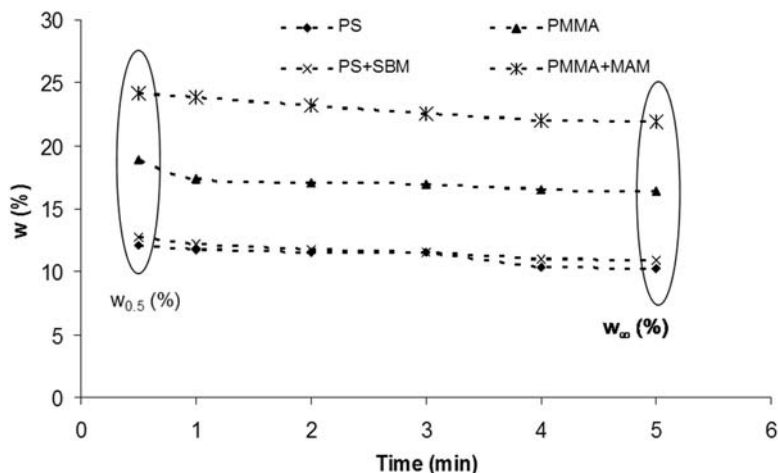


Figure 1. Characteristic curves for mass loss as a function of time, measured at room temperature out of the vessel in the samples analysed after a scCO₂ saturation at 40°C, 24h.

After the saturation process, foaming was carried out by depressurization at a constant rate of 60 bar/min. Samples were after removed from the vessel when reaching room temperature and kept 30 min before preparation for SEM observations. Cellular structure was characterized by means of Scanning Electron Microscopy (model HITACHI S-3000N), on liquid nitrogen fractured samples.

Results

Absorption of CO₂

The absorption kinetics of CO₂ was measured experimentally at different temperatures. Experimental data were obtained by removing the samples and immediately transferring them to a high precision balance to record the increase in weight due to gas absorption. In the absence of a magnetic suspension balance to record directly the weight gain ratio of CO₂ inside the reactor, the next experimental procedure was employed. Samples were removed from the vessel and the mass was monitored at room temperature, carrying out several measurements at different times to evaluate the gas loss due to diffusion process. This procedure was employed for all the samples, from all the saturation temperatures. In Figure 1 we present, as an example, the mass evolution of one sample (out of the CO₂ vessel) as a function of time, after a saturation at 40 °C and a saturation time of 24 h. This curve was typically observed in all the materials considered.

It is observed in Figure 1 the mass evolution of the sample as a function of time. It is clear that after 5 min, the mass of all the samples reaches a constant minimum, which indicates that all the CO₂ has escaped out of the sample, and a 'pseudo equilibrium' situation is reached (this situation is not the polymer–gas equilibrium

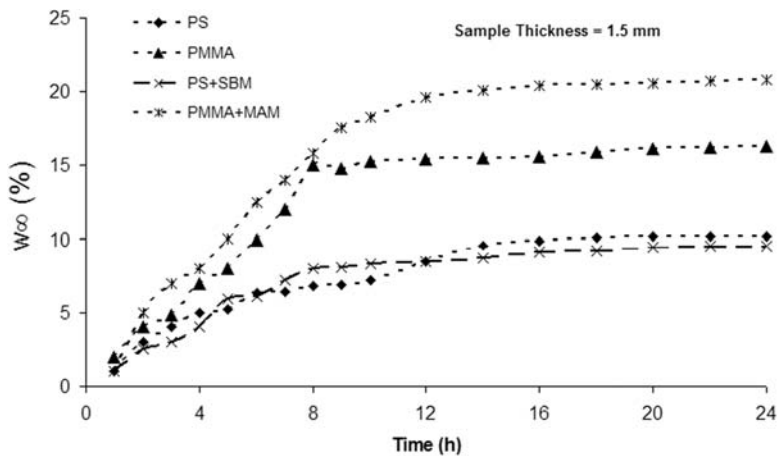


Figure 2. Constant values of CO₂ mass uptake measured at RT, out of the vessel, after scCO₂ exposure at different times (saturation values, 300 bar, 40 °C).

in scCO₂ at the temperature of saturation under pressure). For our calculation purposes, a couple of parameters have been defined as follows: $w_{0.5}$ (%) and w_{∞} (%) represent the weight gain of CO₂ in the first measure (at $t=0.5$ min) and at the pseudo-equilibrium state (at $t\geq 5$ min). Several measurements were performed at higher times, but no differences were found after 5 min from removing the samples from the vessel. We can define an experimental error from the data represented in Figure 1 as follows:

$$\Delta w = \frac{w_{\infty} - w_{0.5}}{w_{0.5}} \tag{1}$$

As explained before, this procedure was followed for all the saturation temperatures, and for all the saturation times up to 24 h. In Figure 2 we present the resultant saturation curves for the materials under study at 40 °C, representing the w_{∞} (%) as a function of saturation time.

The maximum quantity of CO₂ uptake w_{∞} is quantifying the affinity of CO₂ for the material considered; this is the first thermodynamic parameter to be considered for a gas foaming process. This quantity plays a role on the final foam density and on the plasticization of the polymer, lowering the T_g of the polymer due to CO₂swelling, which in turn delays the vitrification of the polymer during foaming, thus allowing more time for cell growth. PMMA and PMMA-based copolymers are, as expected, among the most CO₂-philic polymers in the list chosen. As it can be seen in Figure 2, a constant state in the gas–polymer systems is considered to be reached after 16 h of saturation (300 bar, 40 °C) (the residual mass of CO₂ into the samples at room temperature after saturation at a fixed temperature of 40 °C). It is clear that the measured constant values of mass uptake are not the equilibrium values under CO₂ pressure at various temperatures, but these values are used to estimate the depressed

Table 2. Weight gain ratio of CO₂ at different temperatures

Material	w _∞ (25 °C) (%)	w _∞ (40 °C) (%)	w _∞ (50 °C) (%)	w _∞ (60 °C) (%)	w _∞ (70 °C) (%)	w _∞ (80 °C) (%)
PS	10.4	10.3	10.1	9.4	9.2	8.9
PMMA	16.9	16.4	16.2	16.1	15.9	15.6
PS + SBM	12.1	10.9	10.5	10.4	10.2	9.7
PMMA + MAM	22.3	21.2	20.4	20.1	19.3	19.2

T_g , which nevertheless reveals close to T_g values calculated by other authors. In order to obtain the behavior of the weight gain of CO₂ into the polymer, saturation experiments were carried out at different temperatures, in a similar way as showed in Figure 2. As no great differences were found between 16 and 24 h in the mass uptake measured, samples were saturated at 300 bar during 16 h at temperatures of 25, 40, 50, 60, 70 and 80°C. The results for the weight gain ratio w_{∞} obtained are presented in Table 2, for given saturation conditions of 300 bar and 16 h.

As it can be seen from the data in Table 2, weight gain ratio w decreases as increase in saturation temperature. This result is in accordance with literature^{11,12}, which predicts a classical diminution of the solubility of CO₂ into any polymer as increase in temperature. On the other hand, to analyze the plasticization effect, several theoretical models have been proposed. A simple (due to the accessible material inputs such as molar mass and density) expression was developed by Hwang et al.¹³:

$$T_g = T_{g0} \exp \left[- (M_p)^{-1/3} (\rho)^{-1/4} \alpha \cdot w \right] \quad (2)$$

where α is a material constant depending on both polymer and gas, M_p is the weight average molar mass of the polymer, w is the weight fraction of CO₂ absorbed during the saturation process, T_{g0} is the glass transition temperature of the polymer and ρ_s is the specific density of the polymer.

Hwang's model was successfully employed in our previous works,^{14,15} taking α value as 0.7 and using the values of T_{g0} and M_p obtained experimentally. In this work, we refine the procedure by introducing the experimental error Δ in the calculations. Following this procedure, in Figure 3 a range of depressed glass transition temperatures are given for all the saturation temperatures under study. This estimation has an importance to correlate the solubility of CO₂ into the polymer to the plasticization effect and its effect in the foaming behavior of the microcellular materials produced; i.e. the greater plasticization is, the longer delay for vitrification and the longer growth is allowed to cells.

In our particular case the estimations of Hwang's model can be considered acceptable, and can be correlated to other experimental results presented, by example, by Gengrond et al.¹⁶. As an experimental evidence, we present in Figure 4 two photographs of PS samples, saturated at 40 and 50°C, respectively. It is clear that the sample saturated at 50 °C presents a high deformation due to swelling, whereas

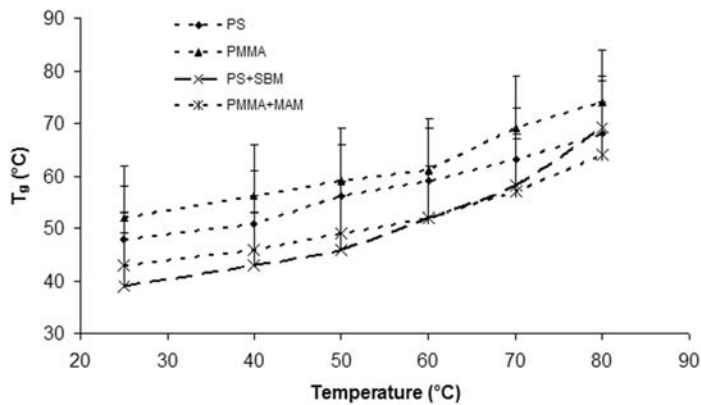


Figure 3. Depressed glass transition temperature ranges estimated after CO₂ absorption at different temperatures.

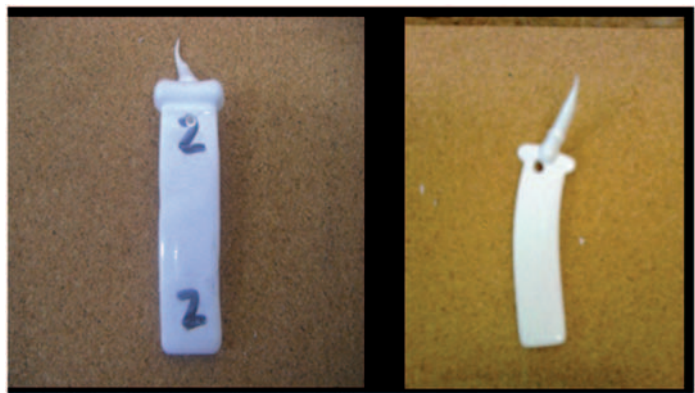


Figure 4. Swelling of two polystyrene samples after saturation at 40 °C and 50 °C, respectively.

deformation in sample saturated at 40 °C is much lower. This indicates that, for the saturation conditions of 300 bar and 16 h, the depressed glass transition temperature lies in the range of 40–50 °C, which is in good agreement with Hwang’s predictions.

Microcellular foaming

Neat homopolymer foams Figure 5 presents the SEM micrographs showing the foamed structures of neat PMMA after two saturation temperatures ($T_{sat}=25$ °C or $T_{sat}=80$ °C). It can be seen that bigger cell sizes are obtained at higher saturation temperatures, due to the larger plasticization interval of the polymer–gas system (i.e. the difference between saturation temperature and estimated T_g).

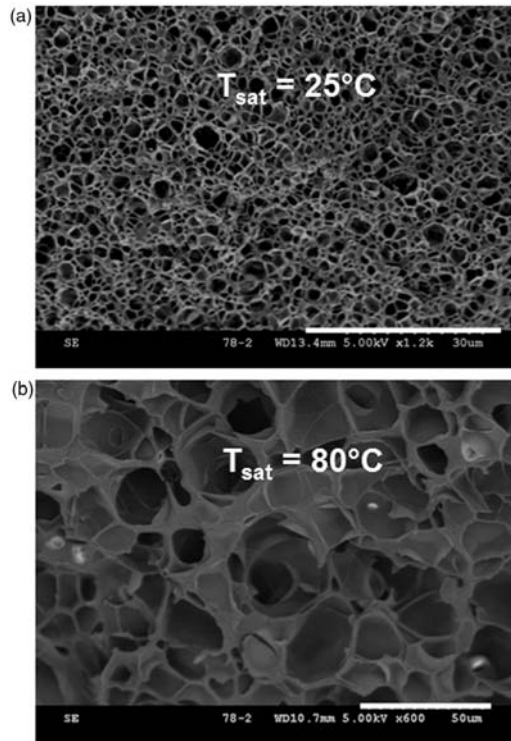


Figure 5. (a) SEM micrographs showing the cell size evolution in PMMA foams Bar scale: 30 μm (b) SEM micrographs showing the cell size evolution in PMMA foams Bar scale: 50 μm .

Therefore at higher saturation temperatures, foaming occurs well above the depressed glass transition temperature in the rubbery state, and the average cell size classically increases with saturation temperature. Moreover, it is important to remark that microcellular foaming is possible even at temperatures below the depressed $T_{g25^\circ\text{C}}$ (e.g. at 52 $^\circ\text{C}$ for the PMMA neat homopolymer), which means that CO_2 is able to expand inside the polymer at a very low degree of plasticization. It is a case of quasi solid state foaming.

On the other side, in Figure 6 are presented two different micrographs showing the cellular structure of neat PS foams obtained after saturation at 25 and 80 $^\circ\text{C}$. Compared to PMMA, the change to PS homopolymer is characterized by a decrease in the affinity for CO_2 , as seen in Table 2, and thus a much greater difficulty for foaming in a one-step CO_2 process. In this case, foaming begins only at higher saturation temperatures (above 60 $^\circ\text{C}$), i.e. when the foaming temperature approaches to depressed glass transition temperature $T_{g25^\circ\text{C}} = 59^\circ\text{C}$.

To conclude this analysis, Figure 7 presents the variation of average cell size ϕ with saturation temperature for both PS and PMMA foams. As it is shown in Figure 7, foaming of PS is only possible above 50 $^\circ\text{C}$, and average cell sizes are

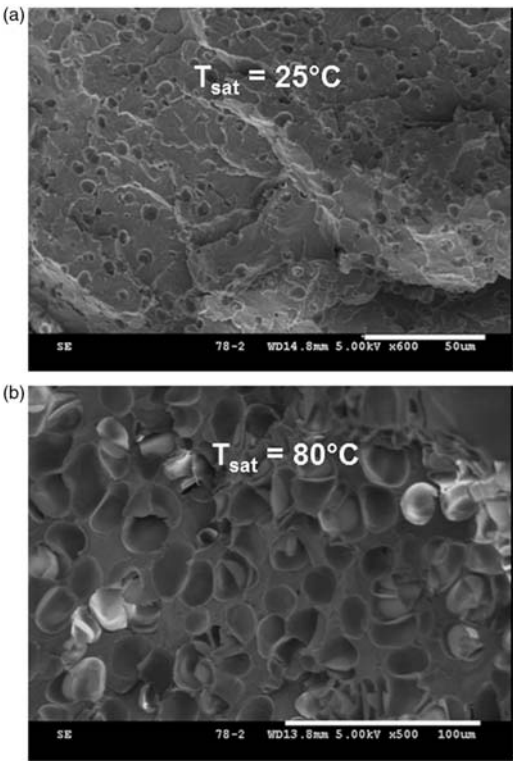


Figure 6. (a) SEM Micrographs showing the cell size evolution in neat PS foams Bar scale: 50 μm (b) SEM Micrographs showing the cell size evolution in neat PS foams Bar scale: 100 μm .

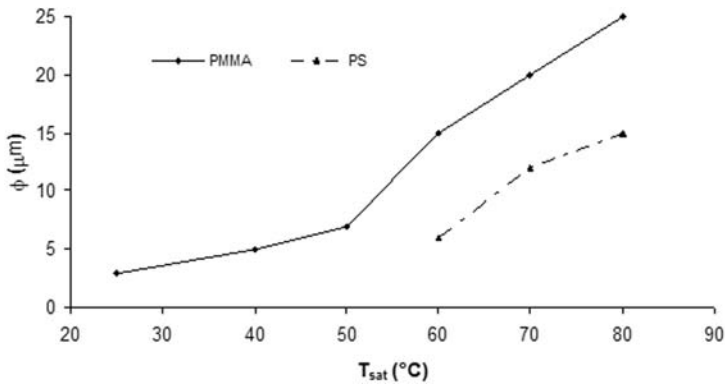


Figure 7. Dependence of the average cell size with saturation temperature in PS and PMMA foams.

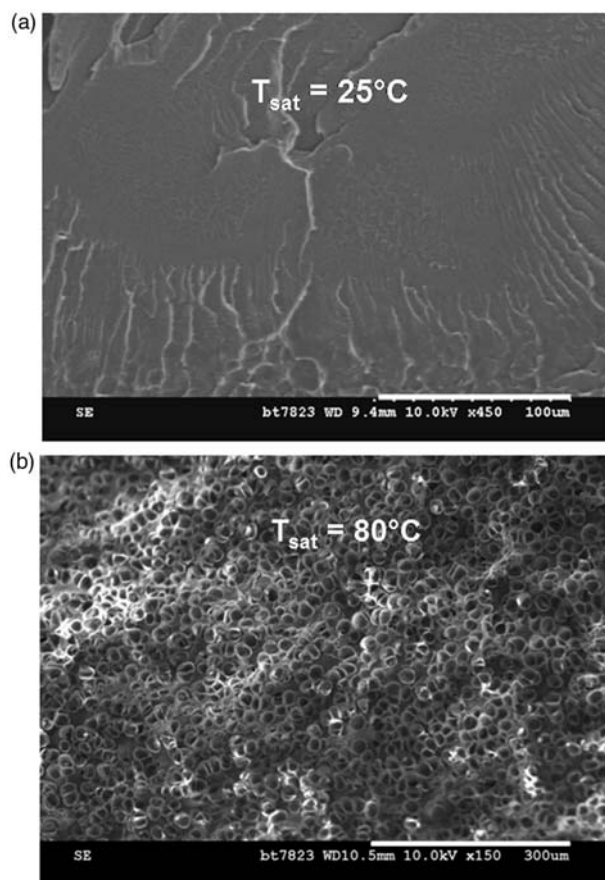


Figure 8. (a) SEM micrographs showing the cell size evolution of the PS;+SBM blend Bar scale: 100 μm ; (b) SEM micrographs showing the cell size evolution of the PS;+SBM blend Bar scale: 300 μm .

ranging between 5 and 15 μm . For PMMA, foaming is possible for the entire temperature range, with a wider cell size range, from 2 to 25 μm . This is again related to the higher CO_2 affinity of the PMMA compared to PS, which allows the production of microcellular foams for the temperature range.

Polymer blend foams Concerning the blends, Figure 8 shows the cellular structure of foams based on PS+SBM (90/10 wt%). This blend presents a more or less comparable behavior to the neat PS foam. Indeed, even if the SBM copolymer is dispersed at a nano-meter level it does not enhance greatly the CO_2 initial absorption (see Table 2). For this reason, PS+SBM blends only foam at temperatures above 50°C . These results are in accordance with the behaviour of the neat PS showed in Figure 6 and the addition of the triblock SBM does not improve the cellular structure at low foaming temperatures. However, increasing the process

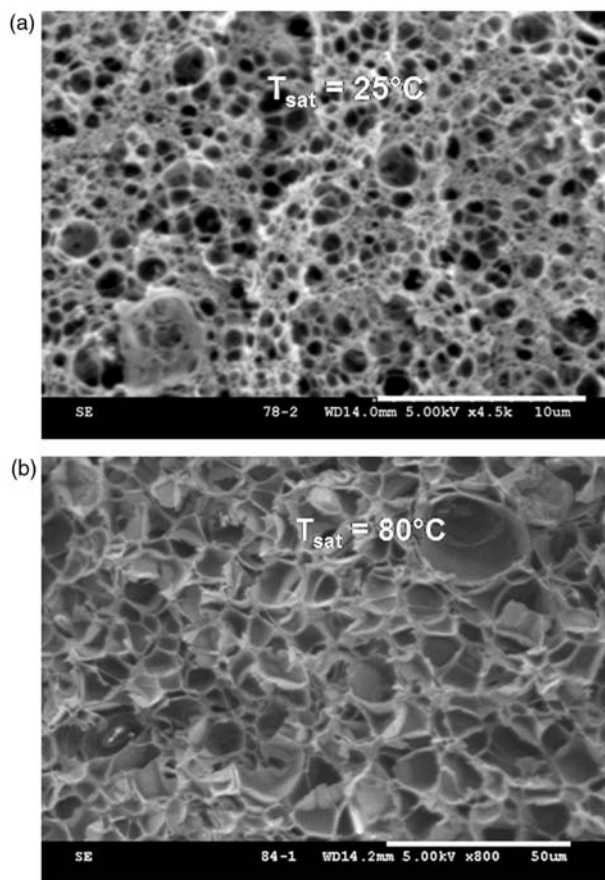


Figure 9. (a) SEM micrographs showing the cell size evolution of the PMMA;+ MAM blend Bar scale: 10 μm ; (b) SEM micrographs showing the cell size evolution of the PMMA;+ MAM blend Bar scale: 50 μm .

temperature up to 80°C results in a very homogeneous cellular structure, (see Figure 8(b)), with a lower average cell size of 30 μm and a higher number of nucleated sites than in neat PS foam. The role of a block copolymer additive is here evidenced.

Figure 9 presents two SEM micrographs showing the cellular structure of the PMMA + MAM (90/10 wt%) foams produced at 25 and 80°C. MAM provides both a higher CO_2 -philicity than PMMA and a rubbery butyl acrylate block. Thus a completely different foaming behavior is obtained. The MAM-based blends first derive a great increase in the CO_2 absorption (see Table 2), and second, they produce ultra-microcellular structures at low saturation temperatures. Again as already noticed, increasing saturation temperatures leads to a large plasticization effect which allows the gas to expand inside the material, producing cells with a higher average size.

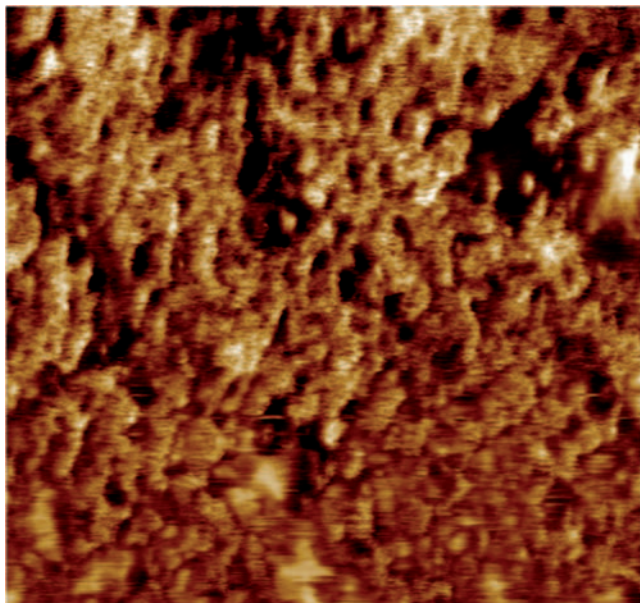


Figure 10. – AFM image of the PMMA+MAM blend (size $1 \times 1 \mu\text{m}$).

Finally this can be also related to the nano-structuration of the MAM copolymer in the PMMA matrix, which is shown in the AFM image obtained from the solid unfoamed material, in Figure 10. The PMMA + MAM blend exhibits worm-like micelles which are CO_2 concentrators and improves the foaming behavior compared to neat PMMA foam.

Figure 11 shows the variation of average cell size ϕ with saturation temperature for PS + SBM and PMMA + MAM foams.

From the previous figure, it is again evidenced that foaming of PS + SBM blends is only possible above 60°C , and no remarkable improvements respect to neat PS foams are observed. However, in the case of PMMA + MAM foams, a noticeable reduction in the average cell size is obtained respect to neat PMMA foams. In this case, average cell size lies between 1 and $8 \mu\text{m}$ for all the foaming temperatures, producing ultra-microcellular foam materials.

To conclude the study, Table 3 presents the foam densities for all the materials investigated. Density varies between 1.13 and 0.65 g/cm^3 , and a maximum expansions rate of to 1.8.

Conclusions

In this work, microcellular foaming of amorphous polymers, polymethylmethacrylate and polystyrene, was studied in supercritical CO_2 via a single-step in the presence of two types of additives which were able to change their foaming behavior and therefore the porous structures. In the first part of the work, the CO_2 saturation

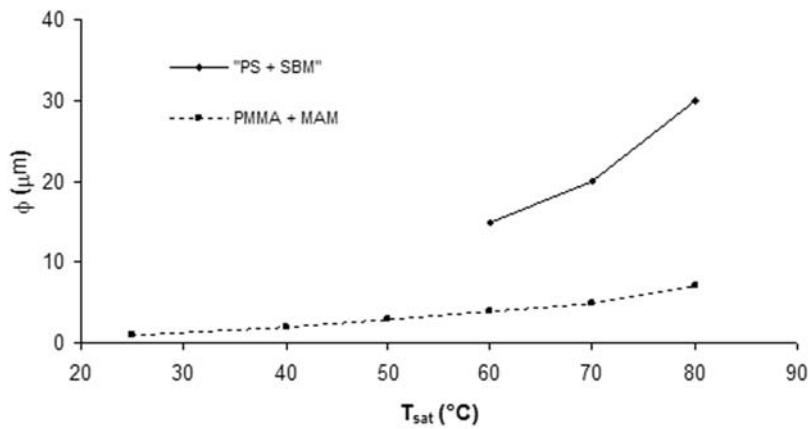


Figure 11. Dependence of the average cell size with saturation temperature in PS+SBM and PMMA+MAM foams.

Table 3. Foam densities obtained of neat homopolymer and homopolymer/copolymer blend foams

Material	ρ25° (g/cm3)	ρ40° (g/cm3)	ρ50° (g/cm3)	ρ60° (g/cm3)	ρ70° (g/cm3)	ρ80° (g/cm3)
PS	NF	NF	1.05	0.94	0.84	0.70
PMMA	1.13	1.05	0.97	0.88	0.83	0.81
PS + SBM	NF	NF	10.5	0.94	0.85	0.76
PMMA + MAM	1.00	0.93	0.92	0.83	0.78	0.65

curves of the materials under study were analysed, showing that methacrylate based polymers and blends presented the highest CO₂ affinity, with a weight gain ratio close to 20 wt%. On the contrary, styrene based polymers presented lower CO₂ absorption values, about 10 wt%. Next, a batch of single-step foaming process was employed to obtain foams from a fixed depressurization rate (60 bar/min) but with different saturation temperatures, from 25 to 80°C. Then we studied the effect of block copolymer additives, some of them are able to enhance greatly the CO₂ uptake (MAM copolymer) and change the foaming behavior, especially the average cell size. Playing on the CO₂ philicity and the solubility of CO₂ in the polymer, the vitrification of the polymer matrix is modified, together with the resulting plasticization delay, i.e. the period or temperature range (δT) in which the polymer gas system remains rubbery.

Concerning the effect of the copolymer additives in the foaming behavior, only a small improvement is observed at high temperatures in the case of adding SBM to PS matrix due to a weak CO₂ solubility difference. However, the presence of well dispersed SBM additive brings an improvement in the cell size and foam homogeneity. If the additive is highly CO₂-philic, as a MAM copolymer, a drastic reduction

of the average cell size is observed, producing ultra-microcellular foams with an average cell size close to 0.1 μm . In this latter case, the CO_2 -philic dispersed entities (MAM block copolymer) but also the always rubbery butyl acrylate block provides a good compromise between CO_2 phicity, nano-sized dispersion of the additive, and possibly the cell growth limitation within the additive (due to being surrounded by the other glassy blocks).

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